



**Verified Carbon
Standard**

AKSARAY LANDFILL GAS POWER PLANT PROJECT

Document Prepared by GAIA Climate



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Aksaray Solid Waste Management System and Electricity Generation Facility (hereafter named as “the Aksaray LFG”) (1.414 MWe) is situated in the province of Aksaray, in the Merkez District, and is owned and operated by Biotrend. The facility is designed to process municipal waste from Aksaray province, classified as type-2 sanitary landfill, and serves a total of 8 districts. The facility consists of four key areas: a sanitary landfill, a mechanical separation facility, a compost facility, and an electricity generation facility.

Upon arrival, mixed municipal waste is sorted at the Mechanical Separation Facility, with biodegradable waste transported to the Compost Facility for further processing, and non-recyclable waste transported to the sanitary landfill. Recyclable materials are sent to licensed companies. The methane gas extracted from the landfill is used to produce electrical energy at the Electricity Generation Facility, which is then supplied to the interconnected system for commercial use. Plant is connected to grid via Aksaray transformer station, run by Meram Distribution System Operator.

The Aksaray LFG plant has obtained a generation license from EMRA, with an expected annual generation of 42.420.000 kWh/year. The project was commissioned on 26.06.2021, with an installed capacity of 1.414 MWe, however, energy supply to the national grid could start on 03.12.2021.¹ The delay is due to operational requirements. The operation could start only after the necessary level of gas accumulation was obtained. Therefore, Project activity date 03.12.2021 is also the starting of crediting period. The project's first crediting period is set at 7 years, with expected average annual emission reductions of 29.019,74 (tCO₂e) per annum. Through this project, waste is stabilized, waste storage is minimized, electricity is produced, and negative environmental impacts are eliminated.

Audit Type	Period	Program	VVB Name	Number of years
Validation	(03.12.2021 – 02.12.2028)	VCS	Earthood Services Private Limited	7
Total				

1.2 Sectoral Scope and Project Type

The project falls into sectoral scope 01: Energy industries (renewable - / non-renewable sources) (Type I Component), sectoral scope 13: Waste handling and disposal (Type III Component). The project is not a grouped project.

¹ The first invoice by the Project Owner was issued on 15/01/2021 for the electricity supply to the national grid. It will be provided to VVB for justification. The EPIAŞ data justifying that energy could only be supplied to the grid as of 03/12/2021 will also be provided to VVB in CSV format.

1.3 Project Eligibility

The VCS program deems the project eligible, as it fulfills the applicability conditions outlined in ACM0001. This enhances the impact and credibility of the project.

1.4 Project Design

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

Not applicable since the project is not a grouped project.

1.5 Project Proponent

Organization name	BIOTREND Çevre ve Enerji Yatırımları Anonim Şirketi
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1.6 Other Entities Involved in the Project

There are no other entities involved.

1.7 Ownership

MD Biyokütle Enerji Üretim A.Ş., a subsidiary of Biotrend Çevre ve Enerji Yatırımları A.Ş., holds all legal rights to the project as per the generation license granted by the Energy Market Regulatory Authority (EMRA) on April 15, 2021. The license remains valid until the date of 21.08.2045.

1.8 Project Start Date

The start date of the Project Activity is 03.12.2021 which corresponds with the provisional acceptance date.

1.9 Project Crediting Period

The start date of the crediting period of the Project Activity has been taken as 03.12.2021, which is nearly 5 months after than the provisional acceptance date as explained in part 1.1. The end date of the first crediting period is 02.12.2028. Project crediting period will be renewed twice which will make the total crediting period 21 years.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	x
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2021	444,72
2022	13.245,74
2023	19.014,39
2024	24.476,76
2025	29.649,50
2026	34.548,32
2027	39.188,08
2028	42.570,69
Total estimated ERs	203.138,21
Total number of crediting years	7
Average annual ERs	29.019,74

1.11 Description of the Project Activity

The project aims to decrease greenhouse gas emissions by utilizing landfill gas (LFG) for electricity generation. By doing so, the project prevents the release of carbon dioxide (CO₂) from fossil fuel-based power plants and minimizes GHG emissions from LFG. The LFG to electricity project includes an LFG collection system, pre-treatment system, electricity generation system, Composting system and flare system.

The LFG collection system uses a booster system to extract LFG from the landfill, followed by blower units that pressurize the gas to the gas engines. The gas engines then burn the LFG to produce electricity, with any excess gas being burned in the enclosed flare system.

Prior to electricity generation, the LFG is treated to remove impurities and moisture, which can cause corrosion in the engines. This treatment process involves filtration, dewatering, cooling, and pressurization.

The project employs four gas engines, each with a capacity of 1,414 MWe, totaling to 5.656 MWe. Yet, only one of them is active since project start date. The generated power is supplied to the grid and delivered to consumers via power transmission lines.

Technology:

The implementation of an active recovery system composed of advanced landfill gas collection and flaring technology will result in significant improvements.

Number	MACHINE AND EQUIPMENT (Type and Technical Properties)
4	Gas Engine (With generator 1.414 kWh _e)
	Blower set
	Booster Unit
	Flare Unit
	Gas Storage
	Transformer

Technical parameters of the motor engines and generators are given below Table.

Table 1 Technical parameters of the gas engine

Technical Specifications	
Electrical Output	1415 kWe
Voltage	400 V
Thermal Output	1426 kW
Thermal Efficiency	43,3%
Fuel Type	Flexible



On 26.06.2021, the inaugural gas engine was officially commissioned, nearly 5 months ago from project activity starting date.

1.12 Project Location

The Aksaray Solid Waste Management Facility and Electricity Generation Facility project is situated in the Merkez District of Aksaray Province.

The detailed coordinates of the project border are presented in Table.

Point	y	x	Latitude	Longitude
1	580992.39	4235137.02	38.25897121	33.92541673
2	581004.51	4235118.79	38.25880584	33.92555316
3	581005.90	4235111.64	38.25874128	33.92556823
4	581024.22	4235079.55	38.25845046	33.92577393
5	581068.05	4235015.23	38.25786690	33.92626749
6	581183.73	4234868.16	38.25653117	33.92757268
7	581221.69	4234811.50	38.25601716	33.92800000
8	581252.80	4234771.51	38.25565399	33.92835095
9	581281.25	4234741.16	38.25537793	33.92867259

10	581292.49	4234731.07	38.25528599	33.92879989
11	581120.51	4234553.27	38.25369932	33.92681414
12	581231.34	4234441.14	38.25267888	33.92806783
13	581222.31	4234432.02	38.25259751	33.92796360
14	581314.72	4234346.86	38.25182176	33.92900985
15	581332.93	4234330.15	38.25166953	33.92921603
16	581048.98	4234030.53	38.24899522	33.92593694
17	580940.53	4233914.35	38.24795806	33.92468441
18	580907.95	4233876.89	38.24762343	33.92430785
19	580853.53	4233962.02	38.24839545	33.92369571
20	580833.01	4233981.30	38.24857104	33.92346342
21	580795.31	4234022.62	38.24894677	33.92303734
22	580733.52	4234104.48	38.24968999	33.92234058
23	580739.60	4234121.94	38.24984678	33.92241205
24	580751.32	4234166.25	38.25024502	33.92255102
25	580751.32	4234166.25	38.25024502	33.92255102
26	580758.51	4234198.03	38.25053075	33.92263680
27	580775.35	4234245.46	38.25095664	33.92283464
28	580785.27	4234285.03	38.25131233	33.92295251
29	580787.67	4234325.14	38.25167356	33.92298451

30	580823.20	4234448.66	38.25278344	33.92340461
31	580855.15	4234527.30	38.25348921	33.92377870
32	580862.67	4234558.83	38.25377266	33.92386823
33	580868.21	4234588.19	38.25403673	33.92393489
34	580872.71	4234598.03	38.25412500	33.92398744
35	580933.86	4234870.71	38.25657669	33.92471740
36	580950.94	4234886.79	38.25672005	33.92491443
37	580960.78	4234898.59	38.25682550	33.92502823
38	580964.90	4234909.56	38.25692398	33.92507657
39	580965.61	4234918.70	38.25700628	33.92508573
40	580963.24	4234927.18	38.25708291	33.92505961
41	580955.49	4234944.47	38.25723941	33.92497302
42	580952.23	4234957.31	38.25735541	33.92493723
43	580952.84	4234965.98	38.25743348	33.92494519
44	580957.23	4234986.01	38.25761358	33.92499765
45	580964.97	4235037.23	38.25807444	33.92509196
46	580968.16	4235074.11	38.25840649	33.92513263
47	580970.66	4235091.41	38.25856216	33.92516317
48	580974.92	4235108.24	38.25871344	33.92521378

1.13 Conditions Prior to Project Initiation

Before the project activity was initiated, the existing scenario (baseline scenario) was as follows: Landfill site in Aksaray released LFG into the atmosphere without any flaring process.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The following laws and regulations regarding the existing and planned activities of the facility will be complied with;

- Fisheries Law No. 1380
- Law No. 1595 on the Establishment and Duties of the Ministry of Forestry
- Law No. 167 on Groundwater
- Law No. 2863 on the Protection of Cultural and Natural Assets
- Environmental Law No. 2872
- National Parks Law No. 2873
- Highway Traffic Law No. 2918
- Law No. 3996 on the Making of Some Investments and Services within the Framework of the Build-Operate-Transfer Model
- Law No. 4126 on the Improvement of Olive Cultivation and the Vaccination of Wilds
- Pasture Law No. 4342
- Tender Law No. 4734
- Labor Law No. 4857
- Land Hunting Law No. 4915
- Metropolitan Municipality Law No. 5216
- Public Health Law No. 593
- Occupational Health and Safety Law No. 6331
- Forest Law No. 6831
- Highway Traffic Regulation, which came into force after being published in the Official Gazette dated 18/07/1997 and numbered 23053
- Occupational Health and Safety Regulation, published in the Official Gazette dated 09/12/2003 and numbered 25311
- Regulation on Control of Excavated Soil, Construction and Demolition Wastes, which entered into force by being published in the Official Gazette dated 18/03/2004 and numbered 25406
- Regulation on Control of Waste Batteries and Accumulators, published in the Official Gazette dated 31/08/2004 and numbered 25569
- Regulation on Wildlife Protection and Wildlife Development Areas, which entered into force after being published in the Official Gazette dated 08.11.2004 and numbered 25637

- Regulation on the Establishment, Management and Supervision of Hunting and Wild Animal Breeding Places and Stations and Rescue Centers, which was published in the Official Gazette dated 30/11/2004 and numbered 25656
- Water Pollution Control Regulation, which entered into force after being published in the Official Gazette dated 31/12/2004 and numbered 25687
- Regulation on the Usage Procedures and Principles of Biocidal Products, which entered into force by being published in the Official Gazette dated 27/01/2005 and numbered 25709
- Regulation on Water Intended for Human Consumption, which entered into force after being published in the Official Gazette dated 17/02/2005 and numbered 25730
- Waste Management Regulation, published in the Official Gazette dated 02/04/2015 and numbered 29314
- Regulation on Business and Work Licenses, which entered into force after being published in the Official Gazette dated 10/08/2005 and numbered 25902
- Regulation on Control of End-of-Life Tires, which was published in the Official Gazette dated 25/11/2006 and numbered 26357
- Regulation on the Structures to be Constructed in Disaster Areas, published in the Official Gazette dated 14/07/2007 and numbered 26582
- Regulation on Buildings to be Constructed in Earthquake Zones, published in the Official Gazette dated 06/03/2007 and numbered 26454
- Air Quality Assessment and Management Regulation, published in the Official Gazette dated 06/06/2008 and numbered 26898
- Waste Oil Management Regulation, published in the Official Gazette dated 21/12/2019 and numbered 30985
- Regulation Amending the Air Quality Assessment and Management Regulation, which came into force after being published in the Official Gazette dated 05/05/2009 and numbered 27219
- Regulation on Control of Industrial Air Pollution, which entered into force by being published in the Official Gazette dated 03/07/2009 and numbered 27277
- Regulation on Control of Medical Wastes, which entered into force by being published in the Official Gazette dated 25/01/2017 and numbered 29959
- Regulation on Landfilling of Wastes, which entered into force by being published in the Official Gazette dated 26/03/2010 and numbered 27533
- Regulation on Evaluation and Management of Environmental Noise, published in the Official Gazette dated 04/06/2010 and numbered 27601
- Regulation on Control of Soil Pollution and Point Source Contaminated Fields, which entered into force by being published in the Official Gazette dated 08/06/2010 and numbered 27605
- Regulation on Incineration of Wastes, which entered into force after being published in the Official Gazette dated 06/10/2010 and numbered 27721
- Communiqué on the Recovery of Certain Non-Hazardous Wastes, which was published in the Official Gazette dated 17/06/2011 and numbered 27967

- Packaging Waste Control Regulation, which entered into force by being published in the Official Gazette No. 27/12/2017 and No. 30283
- Regulation on the Protection of Groundwaters Against Pollution and Deterioration, which entered into force by being published in the Official Gazette dated 07/04/2012 and numbered 28257
- Regulation on the Quality of Surface Waters from which Drinking Water is Obtained or Planned to be Obtained, dated 29/06/2012 and numbered 28338
- Regulation on the Protection of Water Basins and Preparation of Management Plans, which entered into force after being published in the Official Gazette dated 17/10/2012 and numbered 28444
- Regulation on the Control of Odor-Creating Emissions published in the Official Gazette dated 19/07/2013 and numbered 28712
- Exhaust Gas Emission Control Regulation, published in the Official Gazette dated 11/03/2017 and numbered 30004
- Wetlands Protection Regulation, which entered into force by being published in the Official Gazette dated 04/04/2014 and numbered 28962
- Regulation on the Monitoring of Greenhouse Gas Emissions, published in the Official Gazette dated 17/05/2014 and numbered 29003
- Communiqué on Waste Derived Fuel, Additional Fuel and Alternative Raw Materials published in the Official Gazette dated 20/06/2014 and numbered 29036
- Environmental Permit and License Regulation, published in the Official Gazette dated 10/09/2014 and numbered 29115
- Compost Communiqué published in the Official Gazette dated 05/03/2015 and numbered 29286
- Waste Management Regulation published in the Official Gazette dated 02/04/2015 and numbered 29314
- The Regulation on Control of Waste Vegetable Oils, which entered into force after being published in the Official Gazette dated 06/06/2015 and numbered 29378
- Zero Waste Regulation published in the Official Gazette dated 12/07/2019 and numbered 30829
- Waste Electrical and Electronic Equipment Control Regulation, published in the Official Gazette dated 22/05/2012 and numbered 28300
- The Regulation on Flood and Dirt Control, which entered into force by being published in the Official Gazette dated 03/05/2019 and numbered 30763

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is neither registered nor pursuing registration under any other GHG programs

1.15.2 Projects Rejected by Other GHG Programs

No other GHG programs have rejected the project.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services² posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with [organization name(s)] [name of good or service]” since the project’s start date?

☐ Yes

☒ No

There are no relevant legislation in Turkey which enforce such a public statement therefore this is not practicable.

Has the project proponent posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with [name of good or service][describe the region or location, including organization name(s), where practicable].”

☐ Yes

☒ No

In Turkey, the transparency platform of EPIAS doesn’t provide public statements regarding VCU status of reduction and removal projects. Therefore this is not practicable.

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☐ Yes

☒ No

In this project, the produced good is electricity which falls under scope 1 and 2 emissions of industrial producers. There is one national grid in Turkey and there is no potential risk of double counting.

1.17 Sustainable Development Contributions

² Impacted goods and services are all goods and services directly impacted by the technologies and measures specified as project activities in the project description. Please see the VCS Program document *VCS Program Definitions* for additional information.

The sustainable development impact of the Project is expected to be substantial in several ways. Firstly, it will increase the demand for skilled labor in the fabrication, installation, operation, and maintenance of the methane recovery and electricity generation system, leading to sustainable economic growth in the area. Secondly, the generation and delivery of electricity from a renewable and sustainable energy source to a grid that currently depends on fossil fuels will contribute to the fight against climate change. Thirdly, by reducing CH₄ emissions, the project will further support global efforts to combat climate change. Fourthly, the project will introduce a new, clean, and efficient technology model for waste disposal and handling, setting an example for others to follow. Finally, the project will enhance air quality in the local environment by reducing odor, having a positive impact on the community's health and well-being. Nevertheless, it is evident that the project will significantly advance progress towards at least three Sustainable Development Goals (SDGs).

SDG Goal 7: Affordable and Clean Energy

The waste to power project proposal aims to generate renewable energy by capturing methane from municipal waste and utilizing it to produce thermal and electric energy through gas engine systems. The energy generated will be supplied to the national grid, thus increasing the proportion of renewable energy in the global energy mix. Additionally, it will contribute to the increased reliance on clean fuels and technology for the population. This project is expected to have a significant impact in promoting sustainability and reducing the carbon footprint.

SDG Goal 8: Decent Work and Economic Growth

The demand for food and electricity in Turkey is on the rise due to industrialization, urbanization, economic development, and population growth. However, the country's agricultural production capacity is dwindling, and it cannot keep up with the increasing demand. This has resulted in a heavier reliance on imported agricultural products. The reasons for the decreasing domestic agricultural production capacity include the shrinking of agricultural land, rural-urban migration, and rising input costs.

Moreover, Turkey's conventional resources for generating energy, such as coal, oil, and natural gas, are insufficient to meet the growing electricity demand. As a result, the country has become increasingly dependent on foreign energy sources.

To address this issue, a proposed project aims to generate renewable energy from municipal waste. This will reduce Turkey's reliance on foreign energy and achieve higher levels of economic productivity. Additionally, the project will increase employment opportunities in the region, leading to a decrease in unemployment rates. Overall, this project will have a positive impact on Turkey's economy and energy independence.

SDG Goal 13: Climate Action

The upcoming project activity aims to curb greenhouse gas emissions by capturing and utilizing methane, which is one of the most powerful contributors to climate change. The project is projected to yield an average annual emission reduction of approximately 29.019,74 (tCO₂e).

1.18 Additional Information Relevant to the Project

Leakage Management

Leakage emissions are considered as per the applicable tool.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

The Ministry of Environment and Urbanization has granted authorization for the Environmental Impact Assessment (EIA) Report in 25.05.2021, adhering to strict Turkish regulations that require any project to have a neutral or beneficial impact on both the environment and economy. To ensure the project meets these criteria, all necessary measures will be taken to minimize the adverse effects, including but not limited to odor, dust, and gas emissions, as outlined in the comprehensive Environmental Impact Assessment Report.

2.2 Local Stakeholder Consultation

The project's local stakeholder consultation meeting taken place on November 9th March, 2021.

The meeting was held on 09.03.2021 in the conference hall of Aksaray municipality, under the chairmanship of İdris Demirbas, the branch manager of EIA permits in Aksaray provincial directorate of environment and urban planning. Meeting held in a near place to project location in the area of the existing Solid Waste Disposal Facility in Aksaray Province, Merkez district. Meeting started at 14:00 in the under the chairmanship of EIA and Permission director.

Before the public participation meeting, the announcements stating the date, time, place and subject of the meeting were published in the Aksaray provincial Directorate of Environment and City Office bulletin board local level on between 23.02.2021 and 03.03.2021. Also, time and place has been announced via governship of Aksaray between 20.02.2021 and 08.03.2021 at governship bulletin board.

The project owner aimed to gather advice from local residents who might be affected by the project by distributing questionnaires. The purpose of these questionnaires was to collect opinions regarding the potential impact of the project on the local society, environment, economy, daily life, and other related aspects.

The meeting was attended by 13 stakeholders, including representatives of local residents, solid waste management station, and the project owner. The project owner appointed one person to record the minutes of the meeting.

The age range of the stakeholders was between 25 and 66 years old. Wishes and complains from local people has listened and understood by project investor firm. All constraints have been replied in an explanatory way. Main constraints directed by local people was mainly related to drawbacks and dangers of waste transmission

either smell or solid parts via wind and birds to villages. These constraints have answered in technical and reasonable way to participants.

As a result, during the stakeholder consultancy meeting, some stakeholders expressed concerns about the project's potential negative impacts on employees and local residents' living conditions, such as noise, land occupation, and pollution. The project owner addressed these concerns by offering solutions, such as installing protection precautions for birds and planting green isolation belts to mitigate noise, ensuring that the project site would not occupy farms of residents, recycling waste to reduce emissions, and placing the project site far from the residential area to minimize noise pollution. All stakeholders present at the meeting were satisfied with the proposed measures.

Overall, the stakeholders were pleased with the project's development, as it would have a positive impact on the local economy and increase the income of local residents. Information was also given about the environmental impacts that will occur during the construction and operation phases and the precautions to be taken.

2.3 Environmental Impact

The environmental matters in Turkey are jurisdiction of the Ministry of Environment and Urbanization. The Law defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size and other characteristics. There is an EIA-positive document dated 07.06.2021.

It is, possible to claim that there is no net environmental impact linked to the project.

2.4 Public Comments

On March 9th, 2021, a consultation meeting was held with local stakeholders for the project. This meeting, referred to as the Public Participation Meeting, was conducted under the guidance of the now defunct Aksaray Provincial Directorate of Environment and Urbanization. The venue for the meeting was the existing Solid Waste Disposal Facility located at the Merkez, Aksaray Province.

Armutlu village headman stated that they do not oppose the border demarcation facilities, but they do not want it near their own village since they have questions about truck destinations, and the residents of Merkez district, which is closer to the disposal houses, approached the project positively and did not oppose it.

2.5 AFOLU-Specific Safeguards

Since the project is a non-AFOLU project, this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The consolidated baseline methodology applicable to this project is ACM0001.: Flaring or use of landfill gas--- Version 19.0

Applied tools;

TOOL02 “Combined tool to identify the baseline scenario and demonstrate additionality, Version 07.0”

TOOL03 “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion, Version 03.0”

TOOL04 “Emissions from solid waste disposal sites, Version 08.0”

TOOL05 “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 03.0”

TOOL06 “Project emissions from flaring, Version 04.0”

TOOL08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream, Version 03.0”

3.2 Applicability of Methodology

Applicability Criteria	Justification	Applicable to the Project or not?
The methodology is applicable under the following conditions: (a) Install a new LFG capture system in an existing or new (Greenfield) SWDS where no LFG capture system was or would have been installed prior to the implementation of the project activity; or (b) Make an investment into an existing LFG capture system to increase the recovery rate or change the use of the captured LFG, provided that: (i) The captured LFG was vented or flared and not used prior to the implementation of the project activity; and (ii) In the case of an existing active LFG capture system for which the amount of LFG cannot be collected separately from the project system after the implementation of the project activity and its efficiency is not impacted on by the project system: historical data on the amount of LFG capture and flared is available; (c)	Before the project was initiated, the Aksaray landfill site did not have a system in place to capture LFG, resulting in its direct release into the atmosphere. However, the project aims to remedy this by introducing a new LFG capture system at the site, which will be utilized for power generation. It should be noted that the project will not affect the quantity of organic waste that is typically recycled at the site. Prior to the project, organic waste was discarded at the landfill without any efforts to recycle it.	Applicable

Flare the LFG and/or use the captured LFG in any (combination) of the following ways: (i) Generating electricity; (ii) Generating heat in a boiler, air heater or kiln (brick firing only) or glass melting furnace; and/or (iii) Supplying the LFG to consumers through a natural gas distribution network; (iv) Supplying compressed/liquefied LFG to consumers using trucks; (v) Supplying the LFG to consumers through a dedicated pipeline; (d) Do not reduce the amount of organic waste that would be recycled in the absence of the project activity.		
The methodology is only applicable if the application of the procedure to identify the baseline scenario confirms that the most plausible baseline scenario is: (a) Atmospheric release of the LFG or capture of LFG and destruction through flaring to comply with regulations or contractual requirements, to address safety and odour concerns, or for other reasons; and (b) In the case that the LFG is used in the project activity for generating electricity and/or generating heat in a boiler, air heater, glass melting furnace or kiln: (i) For electricity generation: that electricity would be generated in the grid or in captive fossil fuel fired power plants; and/or (ii) For heat generation: that heat would be generated using fossil fuels in equipment located within the project boundary; (c) In the case of LFG supplied to the end-user(s) through natural gas distribution network, trucks or the dedicated pipeline, the baseline scenario is assumed to be	The project's most viable baseline scenario entails the atmospheric release of LFG and electricity generation through grid-based means.	Applicable

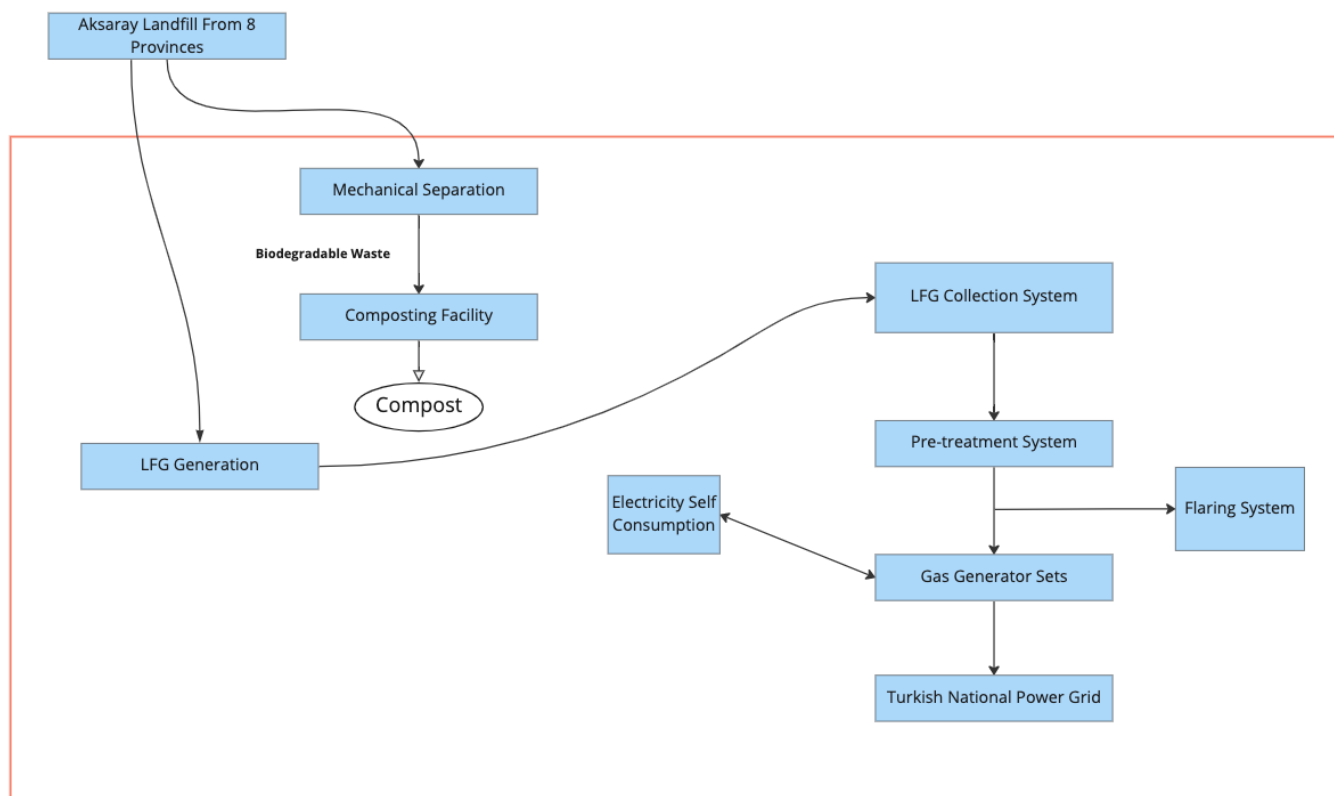
displacement of natural gas. (d) In the case of LFG from a Greenfield SWDS, the identified baseline scenario is atmospheric release of the LFG or capture of LFG in a managed SWDS and destruction through flaring to comply with regulations or contractual requirements, to address safety and odour concerns, or for other reasons.		
This methodology is not applicable: (a) In combination with other approved methodologies. For instance, ACM0001 cannot be used to claim emission reductions for the displacement of fossil fuels in a kiln or glass melting furnace, where the purpose of the CDM project activity is to implement energy efficiency measures at a kiln or glass melting furnace (b) If the management of the SWDS in the project activity is deliberately changed during the crediting in order to increase methane generation compared to the situation prior to the implementation of the project activity.	The project solely utilizes ACM0001. It is crucial to note that the SWDS management will not be altered intentionally during the crediting process to heighten methane production in contrast to the state before project implementation.	N/A

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition	CO ₂	No	The CO ₂ emissions resulting from the organic waste decomposition are not considered, as they are also emitted during the project's implementation.

Source		Gas	Included?	Justification/Explanation
Project	of waste at the SWDS site	CH ₄	Yes	Major source of emissions in the baseline.
		N ₂ O	No	The emission of N ₂ O from solid waste disposal sites (SWDS) is significantly lower in comparison to CH ₄ emissions. This is conservative.
	Emissions from electricity generation	CO ₂	Yes	Major emission source if power generation is included in the project activity
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emissions from electricity consumption due to the Project activity	CO ₂	Yes	May be an important emission source
		CH ₄	No	Excluded for simplification. It is inferred that this source of emission is exceedingly minute.
		N ₂ O	No	Excluded for simplification. It is inferred that this source of emission is exceedingly minute.
	LFG capture and flaring	CO ₂	No	As an integral component of the carbon cycle, this carbon has been previously sequestered from the atmosphere.
		CH ₄	Yes	Emissions are considered negligible
		N ₂ O	No	Emissions are considered negligible

Figure 1 Project Boundary



miro

3.4 Baseline Scenario

The baseline scenario for the Landfill Gas Project entails the decay of biomass and organic matter within the project boundary, releasing methane into the atmosphere in the absence of any project activity. However, any methane emissions that must be removed to comply with national or local safety regulations are excluded from the baseline emissions. In Turkey, since there are no legal requirements for capturing and utilizing LFG emissions, they are considered to be zero for this project.

The recovered methane from the landfill gas is used to generate electricity, which is then exported to the Turkish National Grid. As the national grid primarily relies on thermal power plants that use fossil fuels, the baseline scenario for electricity replacement is determined by multiplying the electricity energy baseline, expressed in kWh of electricity produced by the renewable generating unit, by an emission factor. The emission factor is calculated using the combined margin (CM) approach.

In summary, the baseline emissions for the Landfill Gas Project comprise of two components:

A) the release of methane into the atmosphere as a result of the natural decay of biomass and organic matter, and

B) the electricity produced by the renewable generating unit, which is calculated using the electricity energy baseline and emission factor determined by the combined margin approach.

3.5 Additionality

The project utilizes a simplified procedure to showcase its additionality by employing the powerful methodological tool, "TOOL32: Positive lists of technologies."

As explained above, the project demonstrates that prior to its implementation, the landfill gas (LFG) was merely vented. However, the LFG is now utilized to produce electricity in the project plant with an impressive total nameplate capacity of 5,656 MW. It is essential to note that the capacity is below the 10 MW threshold, making the project automatically additional.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project's baseline emissions scenario involves the substitution of methane from the open-air solid waste disposal site (SWDS) and electricity supplied by the grid. This substitution will be determined using the methodology ACM0001 V.19, ensuring accurate calculations. It comprises the following sources:

- (a) Methane emissions from the SWDS in the absence of the project activity;
- (b) Electricity generation using fossil fuels or supplied by the grid in the absence of the project activity;
- (c) Heat generation using fossil fuels in the absence of the project activity; and
- (d) Natural gas used from the natural gas network in the absence of the project activity.

$$BE_y = BE_{CH_4,y} + BE_{EC,y} + BE_{HG,y} + BE_{NG,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂e/yr)

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO₂e/yr)

$BE_{EC,y}$ = Baseline emissions associated with electricity generation in year y (tCO₂/yr)

$BE_{HG,y}$ = Baseline emissions associated with heat generation in year y (tCO₂/yr)

$BE_{NG,y}$ = Baseline emissions associated with natural gas use in year y (tCO₂/yr)

The project is not involved in heat generation and natural gas use; thus, the baseline emissions is calculated as follows:

$$BE_y = BE_{CH_4,y} + BE_{EC,y}$$

Baseline emissions of methane from the SWDS ($BE_{CH_4,y}$)

The determination of baseline methane emissions from the SWDS involves capturing the methane amount under the project activity and comparing it to the amount that would be captured and eliminated in the absence of the project due to regulations. It also takes into consideration the impact of methane oxidation, which is present in the baseline but absent in the project. This comprehensive approach provides an accurate assessment of the SWDS's methane emissions.

$$BE_{CH_4} = j \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

Where:

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO₂e/yr)

OX_{top_layer} = Fraction of methane in the LFG that would be oxidized in the top layer of the SWDS in the baseline (dimensionless)

$F_{CH_4,PJ,y}$ = Amount of methane in the LFG which is flared and/or used in the project activity in year y (tCH₄/yr)

$F_{CH_4,BL,y}$ = Amount of methane in the LFG that would be flared in the baseline in year y (tCH₄/yr)

GWP_{CH_4} = Global warming potential of CH₄ (tCO₂e/tCH₄)

BE_{CH_4}

Ex post determination of $F_{CH_4,PJ,y}$

Where:

$F_{CH_4,PJ,y}$ = Amount of methane in the LFG which is flared and/or used in the project activity in year y (tCH₄/yr)

$F_{CH_4,flared,y}$ = Amount of methane in the LFG which is destroyed by flaring in year y (tCH₄/yr)

$F_{CH_4,EL,y}$ = Amount of methane in the LFG which is used for electricity generation in year y (tCH₄/yr)

$F_{CH_4,HG,y}$ = Amount of methane in the LFG which is used for heat generation in year y (tCH₄/yr)

$F_{CH_4,NG,y}$ = Amount of methane in the LFG which is sent to the natural gas distribution network and/or dedicated pipeline and/or to the trucks in year y (tCH₄/yr)

The working hours of the gas engines are monitored, and no emission reduction should be claimed for methane destruction during non-working hours.

The project utilizes the LFG only for power generation. Therefore:

$$F_{CH_4,PJ,y} = F_{CH_4,EL,y}$$

$F_{CH_4,EL,y}$ is determined using the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”. The following requirements apply:

(e) As per the gaseous stream tool, if the LFG is used for multiple purposes (e.g. flaring or energy generation), and all methane destruction devices are verified to be operational (e.g. by means of flame detectors records, energy generated), a single flow meter may be used to record the flow into multiple destruction devices. The

destruction efficiency of the least efficient among the destruction devices shall be used as the destruction efficiency for all destruction devices monitored by this flow meter. If there are any periods for which one or more destruction devices are not operational, paragraph 5 (a) and (b) of the Appendix of the "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" tool shall be followed;

(f) CH₄ is the greenhouse gas for which the mass flow should be determined;

(g) The simplification offered for calculating the molecular mass of the gaseous stream is valid (equations (3) or (17) in the tool);

(h) The mass flow should be calculated on an hourly basis for each hour h in year y;

(i) The mass flow calculated for hour h is 0 if the equipment is not working in hour h (Opj,h=not working), the hourly values are then summed to a yearly unit basis.

Amount of the methane used for power generation ($F_{CH_4,EL,y}$) The "Tool to determine the mass of a greenhouse gas in a gaseous stream" provides following 6 options for measuring mass flow of a greenhouse gas i in a gaseous stream Table.

Option	Flow gaseous stream	Volumetric Fraction
A	Volume flow – dry basis	Dry or wet basis
B	Volume flow – wet basis	Dry basis
C	Volume flow – wet basis	Wet basis
D	Mass flow – dry basis	Dry or wet basis
E	Mass flow – wet basis	Dry basis
F	Mass flow – wet basis	Wet basis

Option A is followed since the temperature of the gaseous stream is less than 60°C.

$$F_{i,t} = V_{t,db} \times v_{i,t,db} \times \rho_{i,t}$$

4.2 Project Emissions

In this section all project emission sources are taken into account; assumptions and simplifications are justified. Project emissions are calculated as follows:

$$PE_y = PE_{EC,y} + PE_{FC,y} + PE_{DT,y} + PE_{SP,y} + PE_{FC,j,y} + PE_{flare,y}$$

Where:

PE_y = Project emissions in year y (tCO₂/yr)

$PEEC,y$ = Emissions from consumption of electricity due to the project activity in year y (tCO₂/yr)

$PEFC,y$ = Emissions from consumption of fossil fuels due to the project activity, for purpose other than electricity generation, in year y (tCO₂/yr)

$PEDT,y$ = Emissions from the distribution of compressed/liquefied LFG using trucks, in year y (tCO₂/yr)

$PESP,y$ = Emissions from the supply of LFG to consumers through a dedicated pipeline, in year y (tCO₂/yr)

$PEFC,j,y$ = Are the CO₂ emissions from fossil fuel combustion in process j during the year y (tCO₂/yr)

$PEflare,y$ = Project emissions from flaring of the residual gas in year y

The project does not involve the distribution of compressed/liquefied LFG using trucks nor pipelines. Therefore, $PEDT,y$, $PESP,y$ will be 0. Hence, PEy will be equal to $PEEC,y + PEFC,y$.

$PEEC,y$ is determined by "Tool to calculate baseline, project and/or leakage emissions from electricity consumption" as follow:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y})$$

Where:

$PEEC,y$ = Project emissions from electricity consumption in year y (tCO₂/yr)

$EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y(MWh/yr)

$EF_{EL,j,y}$ = Emission factor for electricity generation for source j in year y(tCO₂/MWh)

$TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y

The "Tool to calculate baseline, project and/or leakage emissions from electricity consumption" offers three distinct scenarios for assessing emissions resulting from various sources of electricity consumption. Scenario A is designed specifically for projects where some of the generated electricity is consumed within the project itself. In such cases, the tool offers two options for calculating the emission factor, allowing for accurate and comprehensive assessments of emissions. For this project, Option A1 ($EF_{EL,j,y} = EF_{grid,CM,y}$) is chosen.

Project emissions from fossil fuel combustion

As per the regulations, a captive power plant or an emergency diesel generator is mandatory to be installed in the event of a shortage of methane for electricity production. This is to ensure the uninterrupted supply of electricity, even in situations where the engines fail to provide the necessary power temporarily. Compliance with this requirement is critical to avoid disruptions to essential services and maintain operational continuity.

CO2 emissions from fossil fuel combustion in process j are calculated based on the quantity of fuels combusted and the CO2 emission coefficient of those fuels, as follows:

$$PE_{FC,j,y} = \sum_j FC_{PJ,j,y} \times COEF_{i,y}$$

Where:

$PE_{FC,j,y}$ = Are the CO2 emissions from fossil fuel combustion in process j during the year y (tCO2/yr)

$FC_{PJ,j,y}$ = Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)

$COEF_{i,y}$ = Is the CO2 emission coefficient of fuel type i in year y (tCO2/mass or volume unit)

I = Are the fuel types combusted in process j during the year y

$$COEF^{33} = 0,00268 \frac{tCO_2}{L} fuel$$

As per tool 3, in order to determine the parameter COEF, option A is the preferred approach as the necessary data is available.

The fuel used in the project is commercial diesel fuel and the quantities are shown below.

The diesel consumption records from after project activity starting date (03.12.2021) are taken into account.

Year	Diesel oil consumption (L,y)	COEF (kgCO2/L)	PE FC (tCO2)
2021	1700	2,68	15,56529315
2022	1700	2,68	195,908
2023	1700	2,68	195,908
2024	1700	2,68	195,908
2025	1700	2,68	195,908
2026	1700	2,68	195,908
2027	1700	2,68	195,908
2028	1700	2,68	180,3427068

Project emissions from flaring

Methodological TOOL06 show the calculation procedure to determine the project emissions from flaring the residual gas PE, flare based on the flare efficiency (flare) and the mass flow of methane to the flare ($F_{CH_4, RG, m}$). The flare efficiency is determined based on monitored data or default values.

In the annex is attached an example of the store data from the monitoring values

The methodological TOOL shows a procedure from three steps to calculate those emissions. The excel tool EB102_repan06_Tool06_Emissions from flare_ was planned to be used to calculate the project emissions from flaring, if working hours wouldn't have been 0.

(a) STEP 1: Determination of the methane mass flow of the residual gas;

(b) STEP 2: Determination of the flare efficiency;

(c) STEP 3: Calculation of project emissions from flaring.

a. STEP 1: Determination of the methane mass flow of the residual gas

The residual gas is monitored continuously in normal conditions (Nm^3/h) every 10 min for 24/7. The mass flow of methane in the residual gas in (kg/m) is calculated

$$F_{CH_4, RG, t} = V_{t, db} \times V_{CH_4, wb, t} \times \rho_{CH_4, n}$$

Where:

$F_{CH_4, RG, t}$	=	Mass flow of greenhouse gas (CH ₄) in the gaseous stream in time interval t (kg gas/h)
$V_{t, wb}$	=	Volumetric flow of the gaseous stream in time interval t on a dry basis (m^3 dry gas/h)
$V_{CH_4, wb, t}$	=	Volumetric fraction of greenhouse gas CH ₄ in the gaseous stream in a time interval t on a dry basis (m^3 gas i/ m^3 dry gas)
$\rho_{CH_4, n}$	=	Density of greenhouse gas CH ₄ in the gaseous stream at normal conditions t (kg gas i/ m^3 gas i)

b. STEP 2: Determination of the flare efficiency;

✓ Enclosed flare

In this case the project implemented an enclosed flare, there are two options to determine the flare efficiency for minute m ($\eta_{flare, m}$)

(a) Option A: Apply a default value for flare efficiency;

(b) Option B: Measure the flare efficiency.

Option A is chosen because:

(a) The temperature of the flare (TEG,m) and the flow rate of the residual gas to the flare (FRG,m) is within the convecos's operating specification for the flare (SPECflare) in the minute m;

- the operating temperature of convecos flare system is 850 – 1100 °C, and according to records temperature were from (0 °C to 40 °C) due to weather conditions meaning not operation.

(b) The flame is detected in the minute m (Flame).

The set of thermocouples, measure continuously temperature and allow the detection of the presence and absence of flame, which can be cross check with the flare working hours and amount of residual gas to flare. T (0) flare is the parameter can be read form the compiled store data from the control system.

Since the enclose flare could be define as low height (8.40 m Heigh) flare the efficiency is adjusted 10 percentile points less form the default value:

η flare,m	80%
----------------	-----

c. STEP 3: Calculation of project emissions from flaring.

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4,RG,m} \times (1 - \eta_{flare,m}) * 10^{-3}$$

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO2e)

GWP_{CH_4} = Global warming potential of methane valid for the commitment period (tCO2e/tCH4)

$F_{CH_4,RG,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ = Flare efficiency in the minute m

Since the commission date the working hours of flare have been in zero.

Therefore:

Flare working hours ³⁶ (h)	0
$PE_{flare,y}$ (tCO2e)	0

$PE_{FC,j,y}$ is determined by “TOOL03 Methodological tool: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion. Version 03.0”. They are the CO2 emissions from fossil fuel combustion in process j during the year y (tCO2/yr).

According to regulation there is to have a captive power plant or urgency diesel generator in case of shortage on methane or a break out. this only implement diesel and has no been used once, please refer to monitoring reports. Therefore, CO2 emissions from fossil fuel combustion in process is taken as zero.

$$PE_{FC,j,y} = 0$$

4.3 Leakage

No leakage effects are accounted for under the ACM0001 Methodology.

4.4 Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂e/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

The baseline emissions are calculated as follows:

$$BE_y = BE_{CH_4,y} + BE_{EC,y}$$

Emissions of methane from the SWDS are calculated as follows:

$$BE_{CH_4} = j \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-kj \times (y-x)} \times (1 - e^{-kj}))$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	464,09	19,37	444,72
2022	13.489,47	243,74	13.245,74
2023	19.258,13	243,74	19.014,39
2024	24.720,50	243,74	24.476,76
2025	29.893,24	243,74	29.649,50
2026	34.792,06	243,74	34.548,32
2027	39.431,82	243,74	39.188,08

2028	42.795,06	224,37	42.570,69
Total	204.844,37	1706,16	203.138,21

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	ϕ										
Data unit	Dimensionless										
Description	Model correction factor to account for model uncertainties										
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”.										
Value applied	0.75										
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: according to the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”. The appropriate factor was taken based on the application of the tool (A). See table below and the climate where the SWDS is located (Dry conditions). <table border="1"> <thead> <tr> <th></th><th>Humid/Wet Conditions</th><th>Dry Conditions</th></tr> </thead> <tbody> <tr> <td>Application A</td><td>0.75</td><td>0.75</td></tr> <tr> <td>Application B</td><td>0.85</td><td>0.80</td></tr> </tbody> </table>			Humid/Wet Conditions	Dry Conditions	Application A	0.75	0.75	Application B	0.85	0.80
	Humid/Wet Conditions	Dry Conditions									
Application A	0.75	0.75									
Application B	0.85	0.80									
Purpose of Data	Calculation of baseline emissions										
Comments	-										

Data / Parameter	OXtop_layer
Data unit	Dimensionless
Description	Fraction of methane that would be oxidized in the top layer of the SWDS in the baseline
Source of data	Consistent with how oxidation is accounted for in the methodological tool “Emissions from solid waste disposal sites” which is based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories.
Value applied	0.1

Justification of choice of data or description of measurement methods and procedures applied	OXtop-layer is the fraction of the methane in the LFG that would oxidize in the top layer of the SWDS in the absence of the project activity. Under the project activity, this effect is reduced as a part of the LFG is captured and does not pass through the top layer of the SWDS. This oxidation effect is also accounted for in the methodological tool “Emissions from solid waste disposal sites”.
Purpose of Data	The oxidation factor shall be included in the calculation of baseline emissions whereas the effect of oxidation is, as a conservative assumption, neglected under the project activity.
Comments	The oxidation factor represents the proportion of methane that is oxidized to CO ₂ . This should be distinguished from the methane correction factor (MCF) which is to account for the situation that ambient air might intrude into the SWDS and prevent methane from being formed in the upper layer of SWDS.

Data / Parameter	F
Data unit	Dimensionless
Description	Volume fraction of methane in the SWDS gas.
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”.
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	The default value 0,5 is recommended by the Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site is applied.
Purpose of Data	Calculation of baseline emissions.
Comments	-

Data / Parameter	MCF
Data unit	-
Description	Methane correction factor.
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”.
Value applied	1
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: according to the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site” and to the IPCC Guidelines for National

	Greenhouse Gas Inventories (2006). The following values are to be considered: -1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e., waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) leveling of the waste; -0.5 for semi-aerobic managed solid waste disposal sites. These must have controlled placement of waste and will include all of the following structures for introducing air to waste layer: (i) permeable cover material; (ii) leachate drainage system; (iii) regulating pondage; and (iv) gas ventilation system; -0.8 for unmanaged solid waste disposal sites ñ deep and/or with high water table. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of greater than or equal to 5 meters and/or high water table at near ground level. Latter situation corresponds to filling inland water, such as pond, river or wetland, by waste; -0.4 for unmanaged-shallow solid waste disposal sites. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of less than 5 metres. Any comment: The methane correct. Due to the fact that the baseline emissions correspond to an unmanaged SWDS. The value taken was 0,8.
Purpose of Data	Calculation of baseline emissions.
Comments	-

Data / Parameter	DOC _f
Data unit	Dimensionless
Description	Fraction of degradable organic carbon that can decompose
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”.
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	The default value 0,5 is recommended by the Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site is applied.
Purpose of Data	Calculation of baseline emissions.
Comments	-

Data / Parameter	DOC _j														
Data unit	Dimensionless														
Description	Fraction of degradable organic carbon (by weight) in the waste type j.														
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”. These data were taken in the environmental assessment report														
Value applied	DOC _j See table below														
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: according to the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”. The following values for the different waste types j are applied <table border="1"> <thead> <tr> <th>Waste Type j</th><th>DOC j (% wet waste)</th></tr> </thead> <tbody> <tr> <td>Wood and wood products</td><td>0</td></tr> <tr> <td>Pulp, paper, and cardboard (other than sludge)</td><td>17,5</td></tr> <tr> <td>Food, food waste, beverages, and tobacco (other than sludge)</td><td>53,5</td></tr> <tr> <td>Textiles</td><td>0</td></tr> <tr> <td>Garden, park, and yard waste</td><td>1,5</td></tr> <tr> <td>Glass, plastic, metal, other inert</td><td>12,4</td></tr> </tbody> </table>	Waste Type j	DOC j (% wet waste)	Wood and wood products	0	Pulp, paper, and cardboard (other than sludge)	17,5	Food, food waste, beverages, and tobacco (other than sludge)	53,5	Textiles	0	Garden, park, and yard waste	1,5	Glass, plastic, metal, other inert	12,4
Waste Type j	DOC j (% wet waste)														
Wood and wood products	0														
Pulp, paper, and cardboard (other than sludge)	17,5														
Food, food waste, beverages, and tobacco (other than sludge)	53,5														
Textiles	0														
Garden, park, and yard waste	1,5														
Glass, plastic, metal, other inert	12,4														
Purpose of Data	Calculation of baseline emissions.														
Comments	-														

Data / Parameter	k		
Data unit	Dimensionless		
Description	Decay rate for the waste type j.		
Source of data	First order decay model from the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site”.		
Value applied	DOCj See table below		
Justification of choice of data or description of measurement methods and	For baseline emissions: according to the “Tool to determine methane emissions avoided from dumping waste at a solid waste disposal site” .The following values for the different waste types j are applied.		
	Type of Waste	Climate Zone	
		Boreal and Temperate	Tropical

procedures applied			(MAT $\leq$ 20 C)		(MAT > 20 C)	
			Dry	Wet	Dry	Moist and Wet
			(MAP/PET < 1)	(MAP/PET > 1)	(MAP < 1000 mm)	(MAP $\geq$ 1000 mm)
			Default	Default	Default	Default
	Slowly degrading waste	Paper/textile waste	0.04	0.06	0.045	0.07
		Wood/straw waste	0.02	0.03	0.025	0.035
	Moderately degrading waste	Other (non-food) organic putrescible/Garden and park waste)	0.05	0.1	0.065	0.17
	Rapidly degrading waste	Food waste/Sewage sludge	0.06	0.185	0.085	0.4
	Bulk Waste		0.05	0.09	0.065	0.17
Purpose of Data	Calculation of baseline emissions.					
Comments	MAT - mean annual temperature, MAP -Mean annual precipitation, PET - potential evapotranspiration. MAP/PET is the ratio between the mean annual precipitation and the potential evapotranspiration. The values applied are for Climate Zone: Boreal and temperate (MAT $\leq$ 20 C), and dry (MAP/PET < 1)					

Data / Parameter	W_{total}	
Data unit	tons	
Description	The amount of waste disposed in the landfill sites in year x	
Source of data	Historical waste data and waste projections	
Value applied	Year	W_{total} (t/y)
	2021	98,260
	2022(January-November)	81,955
Purpose of Data	Calculation of baseline emissions.	
Comments	-	

Data / Parameter	GWP _{CH4}
Data unit	t CO2 e/t CH4
Description	Global warming potential of methane.
Source of data	Decisions under UNFCCC and the Kyoto Protocol (a value of 28 is to be applied)
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions.
Comments	-

Data / Parameter	EF _{CO2,grid, y}
Data unit	t CO2 e/MWh
Description	Emission factor for the Turkish National Grid
Source of data	Published data from The Ministry of the Energy and Natural Resources in Turkey
Value applied	0.552
Justification of choice of data or description of measurement methods and procedures applied	Obtained from the most recent official national grid emission factor data published by the Ministry of the Energy and Natural Resources in Turkey
Purpose of Data	Used in • Calculation of baseline emissions • Calculation of project emissions
Comments	-

Data / Parameter	TDL _{j,y}
Data unit	%
Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Calculated from the Data Provided by Turkish Electricity Transmission Corporation (TEİAŞ) ³³ in accordance with Methodological Tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 3.0
Value applied	0.10768

Justification of choice of data or description of measurement methods and procedures applied	The annual average value based on the most recent data available within Turkey is used from the data provided by TEİAŞ.
Purpose of Data	Used in • Calculation of project emissions
Comments	-

Data / Parameter	$\rho_{CH_4, n}$
Data unit	kg/m ³
Description	Density of methane gas at reference conditions
Source of data	Default Value of TOOL06
Value applied	0.716
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Determination of the methane mass flow of the residual gas
Comments	

Data / Parameter	SPEC _{flare}
Data unit	Temperature - °C Flow rate m ³ /h Maintenance schedule - number of days
Description	Stainless steel burner with multiple nozzle • Combustion range: 25-50% methane • Combustion temperature 850 - 1100 °C • Operating logic managed by PLC • Designed for combustion of 1500 m³/h of biogas. • Flame retention >0.3 sec The operator will decide whether to start and ignite the Flare with Cogeneration either on or off (see settings page on Data sheet specifications).
Source of data	Conveco – specialist in Biogas technology
Value applied	The ones specified on the technical sheet
Justification of choice of data or description of measurement methods and procedures applied	High temperature combustion and extraction system

Purpose of Data	Calculation of project emissions from residual gas
Comments	Please find the specifications on the Folder Data sheet of equipment. This in an enclosed flare.

5.2 Data and Parameters Monitored

Data / Parameter	Management of SWDS
Data unit	-
Description	Management of SWDS
Source of data	Original design of the landfill
Description of measurement methods and procedures to be applied	Project participants should refer to the original design of the landfill to ensure that any practice to increase methane generation have been occurring prior to the implementation of the project activity. Any change in the management of the SWDS after the implementation of the project activity should be justified by referring to technical or regulatory specifications.
Frequency of monitoring/recording	Annually
Value monitored	PO stated that there is no change to the design.
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EG _{PJ,y}
Data unit	MWh
Description	Amount of electricity generated using LFG by the project activity in year y
Source of data	EPIAŞ (Electricity Market Operator Company) records of settlement notifications
Description of measurement methods	Monitor net electricity generation by the project activity using LFG

and procedures to be applied	
Frequency of monitoring/recording	Continuous measurement every 10 min (24/7) all year and data can be downloaded and saved anytime
Value monitored	-
Monitoring equipment	Calibrated electricity meter
QA/QC procedures to be applied	Electricity meter is subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing by regulation to ensure accuracy and TEİAŞ (Transmission System Operator) is responsible of calibrating these meters, which is a government-owned corporation. The readings are double checked by the electricity distribution company. The calibration of meter, including the frequency of calibration, will be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meter is in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements.
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	This parameter is required for calculating baseline emissions associated with electricity generation (BEEC,y) using the methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”

Data / Parameter	EC _{PJ,y}
Data unit	MWh
Description	Amount of electricity consumed by the project activity in year y
Source of data	EPIAŞ (Electricity Market Operator Company) records of settlement notifications
Description of measurement methods and procedures to be applied	Sources of consumption shall include, where applicable, electricity consumed for the operation of the LFG capture system, for any processing and upgrading of the LFG, for transportation of the LFG to the flare or other applications (boilers, power generators), for the compression of the LFG into the natural gas network, etc.
Frequency of monitoring/recording	Continuous measurement and monthly recording
Value monitored	

Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Electricity meter is subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing by regulation to ensure accuracy and TEİAŞ (Transmission System Operator) is responsible of calibrating these meters, which is a government-owned corporation. The readings are double checked by the electricity distribution company. The calibration of meter, including the frequency of calibration, will be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meter is in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements.
Purpose of the data	Calculation of project emissions
Calculation method	-
Comments	This parameter is required for calculating project emissions from electricity consumption due to an alternative waste treatment process t PEEC,y) using the methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”

Data / Parameter	$O_{pj,h}$
Data unit	h
Description	Operation of the equipment that consumes the LFG
Source of data	Project participants
Description of measurement methods and procedures to be applied	For each equipment unit j using the LFG monitor that the plant is operating in hour h by the monitoring any one or more of the following three parameters: (a) Temperature. Determine the location for temperature measurements and minimum operational temperature based on manufacturer’s specifications of the burning equipment. Document and justify the location and minimum threshold in the PDD; (b) Flame. Flame detection system is used to ensure that the equipment is in operation; (c) Products generated. Monitor the generation $O_{pj,h} = 0$ when: (a) One of more temperature measurements are missing below the minimum threshold in hour h (instantaneous measurements are made at least every minute);

	(b) Flame is not detected continuously in hour h (instantaneous measurements are made at least every minute) Otherwise, $Op_{j,h} = 1$
Frequency of monitoring/recording	Hourly
Value monitored	
Monitoring equipment	Flare program logic controller
QA/QC procedures to be applied	-
Purpose of the data	Calculation of baseline emissions
Calculation method	Calculated in the workbook using a formula that checks the flow from the flow meters against the flare temperature and UV sensor and or electricity flow meter.
Comments	None

Data / Parameter	$V_{i,t,wb}$
Data unit	$m^3 \text{ gas } i / m^3 \text{ wet gas}$
Description	Volumetric fraction of greenhouse gas i in a time interval t on a wet basis
Source of data	Measurement
Description of measurement methods and procedures to be applied	Continuous in-situ analyzers
Frequency of monitoring/recording	Continuous measurement and hourly recording
Value monitored	-
Monitoring equipment	Infrared Gas Analyzer The instrument is calibrated regularly utilizing a certified span gas on site until the unit is no longer able to be calibrated on site. Should the instrument not be able to be calibrated then the instrument sensor is replaced.
QA/QC procedures to be applied	

Purpose of the data	Calculation of Baseline Emissions
Calculation method	Not Applicable
Comments	This parameter will be monitored in Option F as per the applied “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (Version 02.0.0).

Data / Parameter	Tariff or electricity exported
Data unit	Currency – Lira
Description	Tariff of electricity exported
Source of data	Power Purchase Agreement
Description of measurement methods and procedures to be applied	Monitored annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years.
Frequency of monitoring/recording	At the first issuance request after each phase of the project is fully implemented
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	Audited by professional, independent financial auditors. The DOE should only verify that the data provided corresponds to the data from independent financial auditors
Purpose of the data	-
Calculation method	-
Comments	The monitoring of this parameter is only required for projects applying the simplified procedures to identify the baseline scenario and demonstrate additionality. Parameter in accordance with the applied ACM0001.

Data / Parameter	$V_{t,wb}$
Data unit	m ³ gas / hours
Description	Volumetric flow of the gaseous stream in time interval t on a wet basis
Source of data	Measurement

Description of measurement methods and procedures to be applied	Volumetric flow measurement should always refer to the actual pressure and temperature. Instruments with recordable electronic signal (analogical or digital) are required
Frequency of monitoring/recording	Continuous measurement and hourly recording
Value monitored	
Monitoring equipment	Flow meter Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration and frequency of calibration is according to manufacturer's specifications.
QA/QC procedures to be applied	-
Purpose of the data	Calculation of Baseline Emissions
Calculation method	Not applicable
Comments	-

Data / Parameter	T
Data unit	°C
Description	Temperature of the gaseous stream in time interval t
Source of data	Measurement
Description of measurement methods and procedures to be applied	Thermocouple
Frequency of monitoring/recording	Continuous measurement and hourly recording
Value monitored	-
Monitoring equipment	Flow meter Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration and frequency of calibration is according to manufacturer's specifications.
QA/QC procedures to be applied	-
Purpose of the data	Calculation of Project Emissions

Calculation method	Not applicable
Comments	-

Data / Parameter	P
Data unit	Pascals (Pa)
Description	Pressure of the gaseous stream in time interval t
Source of data	Measurement
Description of measurement methods and procedures to be applied	Flow meter
Frequency of monitoring/recording	Continuous measurement and hourly recording
Value monitored	-
Monitoring equipment	Flow meter Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration and frequency of calibration is according to manufacturer's specifications.
QA/QC procedures to be applied	-
Purpose of the data	Calculation of Project Emissions
Calculation method	Not applicable
Comments	-

Data / Parameter	$F_{CH_4,EG,t}$
Data unit	kg
Description	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t
Source of data	Measurements undertaken by a third-party accredited entity
Description of measurement methods and procedures to be applied	Values taken from the system that has been calibrated
Frequency of monitoring/recording	Biannual

Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	According to standard applied
Purpose of the data	Calculation of Project Emissions
Calculation method	-
Comments	The project implements an enclosed flare therefore, there is monitoring.

Data / Parameter	$T_{EG,m}$
Data unit	°C
Description	Temperature in the exhaust gas of the enclosed flare in the minute m
Source of data	Project operator's Monitoring reports
Description of measurement methods and procedures to be applied	Values display on the PLC of calibrated instruments
Frequency of monitoring/recording	Once per minute if the flare in is on meaning 100% operation
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	Temperature measurement equipment should be replaced or calibrated in accordance with their maintenance schedule
Purpose of the data	-
Calculation method	-
Comments	The project implements an enclosed flare therefore, there is monitoring.

Data / Parameter	$V_{i,RG,m}$
Data unit	-
Description	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where $i = CH_4, CO_2, O_2, H_2S, N_2$

Source of data	Measurements by project operator using a continuous gas analyzer (values are recorded with the same frequency as the flow).
Description of measurement methods and procedures to be applied	Measurement may be made on wet basis. Analyzers are periodically calibrated according to the manufacturer's recommendation.
Frequency of monitoring/recording	Continuously. Values are to be averaged on a minute basis if flare is on operation 100% status.
Value monitored	-
Monitoring equipment	Biogas Analyzer
QA/QC procedures to be applied	Analyzers must be periodically calibrated according to the manufacturer's recommendation.
Purpose of the data	-
Calculation method	-
Comments	Project operator measure the content CH ₄ , CO ₂ and H ₂ S

Data / Parameter	M _{RG,m}
Data unit	kg
Description	Mass flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	Flare Unit
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital) With calibrated instruments
Frequency of monitoring/recording	Continuous
Value monitored	
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Calibration and frequency of calibration is according to manufacturer's specifications
Purpose of the data	-
Calculation method	
Comments	This parameter is monitored, project implements enclosed flare

Data / Parameter	VO ₂ ,EG,m
Data unit	-
Description	Volumetric fraction of O ₂ in the exhaust gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project operator using a continuous gas analyzer
Description of measurement methods and procedures to be applied	Extractive sampling analyzers with water and particulates removal devices or in situ analyzers for wet basis determination.
Frequency of monitoring/recording	Continuous
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	Analyzers must be periodically calibrated according to the manufacturer's recommendation.
Purpose of the data	-
Calculation method	-
Comments	This parameter is monitored, project implements enclosed flare Equipment works with full combustion

Data / Parameter	f _{CH₄} ,EG,m
Data unit	mg/m ³
Description	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m
Source of data	Measurements by project operator using a continuous gas analyzer
Description of measurement methods and procedures to be applied	Extractive sampling analyzers with water and particulates removal
Frequency of monitoring/recording	Continuously
Value monitored	-
Monitoring equipment	-

QA/QC procedures to be applied	Analyzers must be periodically calibrated according to the manufacturer's recommendation.
Purpose of the data	-
Calculation method	-
Comments	This parameter is monitored, project implements enclosed flare. Since it works with complete combustion, there is no methane left.

Data / Parameter	Flame _m
Data unit	Flame on or Flame off (0% or 100%)
Description	Flame detection of flare in the minute m
Source of data	Measure using a fixed installation optical flame detector with calibrated instrument
Description of measurement methods and procedures to be applied	Once per minute. Detection of flame recorded as a minute that the flame was on.
Frequency of monitoring/recording	Continuously
Value monitored	
Monitoring equipment	-
QA/QC procedures to be applied	Equipment shall be maintained and calibrated in accordance with manufacturer's recommendations
Purpose of the data	-
Calculation method	-
Comments	This parameter is monitored, project implements enclosed flare

Data / Parameter	Maintenance
Data unit	Calendar dates
Description	Maintenance events completed in year y
Source of data	Project operator
Description of measurement methods and procedures to be applied	Record the date that maintenance events were completed in year y. Records of maintenance logs must include all aspects of the maintenance including the details of the person(s) undertaking

	the work, parts replaced, or needing to be replaced, source of replacement parts, serial numbers and calibration certificates
Frequency of monitoring/recording	Annual
Value monitored	-
Monitoring equipment	-
QA/QC procedures to be applied	Records must be kept in a maintenance log for two years beyond the life of the flare
Purpose of the data	-
Calculation method	-
Comments	This parameter is monitored, project implements enclosed flare

5.3 Monitoring Plan

The monitoring system will be managed by Biotrend personnel. The staff members responsible for monitoring tasks will be divided into four categories: Operation Site Manager, Field Operator, Electrical & Mechanical Operator, and Laboratory Operator. The Field Operators, in particular, will perform crucial tasks such as maintaining collection system components, checking blower and flare operation, recording data, monitoring and adjusting LFG extraction wells, and sending information to the Operation Site Manager. They will also be responsible for creating daily logs and completing checklists, ensuring efficient and effective monitoring of the system.

Operation Manager's duties include thoroughly analyzing the data collected by both the Field Technicians and analytical equipment, proposing recommendations, and making necessary system adjustments to optimize the capture and destruction of methane. Additionally, the operation manager will schedule monitoring and O&M activities, conduct quality assurance checks on operations, and coordinate with system component manufacturers to ensure proper operations and calibration. It should be noted that the operation manager will solely be responsible for reviewing manually collected data. Operation Manager's role is pivotal in ensuring optimal system performance and reducing environmental impact.

Project Management Responsibility

The operators of the LFG recovery and electricity generation system will collect all on-site data, while qualified technicians with extensive knowledge of equipment operation, maintenance, and emergency response will manage operations and maintenance. The project manager bears ultimate responsibility for ensuring that all duties are performed correctly, and relevant staff members have been assigned specific roles and responsibilities in VCS monitoring.

Training of Monitoring Personnel

At the project's outset, the monitoring team undergoes training with the primary objective of ensuring seamless project execution. Biotrend will also provide regular and targeted training sessions to further enhance team efficiency.

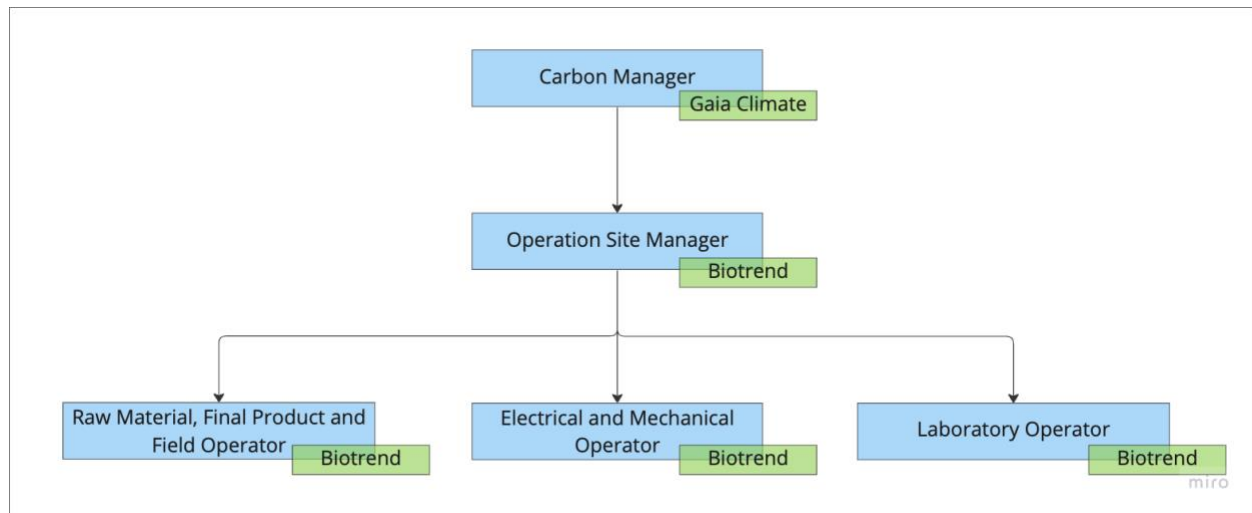


Figure 2 Organizational Structure of the Carbon Project

Data Analysis

The Operation Manager will conduct daily reviews and analyses of the collected data. If any parameter drifts, the Manager or Technician will promptly take action to resolve potential issues. The onsite-monitoring database will contain all necessary data for emission reduction calculations. The Monitoring Manager will ensure the required information is reported to Biotrend.

The following actions are crucial for ensuring the success of the monitoring plan:

Maintenance Plan

- The LFG plant has a computerized management and control system that manages gas meters, heating system, alarm management, and data analysis of LFG analysis for electricity generation and dispatching to the National Grid.
- To ensure proper data monitoring, the system's maintenance requires equipment preventive maintenance and calibration.
- All data collected during monitoring will be securely archived electronically for at least 2 years after the last crediting period.
- Maintenance and calibration of the monitoring equipment for methane recovery will be carried out following the manufacturers' recommendations.
- The high accuracy electricity meter meets state authority's relevant metrological requirements and maintenance procedures will be conducted according to national procedures and standards.

Quality assurance and Control (QA/QC)

Guaranteeing the quality of monitored data is of utmost importance, which is why the project employs rigorous quality control and quality assurance procedures. All data is electronically archived, regularly backed up, and preserved for a minimum of two years following the conclusion of the last crediting period. Any deviations from the permissible parameters trigger prompt corrective actions. The methane recovery equipment is subject to maintenance and calibration in accordance with the manufacturers' recommendations, while the high-precision electricity meter meets all the metrological requirements prescribed by the state authority. The maintenance of the meter will adhere to national procedures and standards

Calibration plans:

MRU BACA GAZI LTD conducts biannual calibration visits for their gas flow meters in the booster unit, which includes measuring O2%, H2S ppm, CO2%, and CH4%. Calibration certificates are available for review. The gas engines are equipped with a monitoring system that displays parameters in real-time. Data is stored internally on a PC and in the cloud, and can easily be downloaded as an Excel file.

Emergencies Procedures

As part of the standard monitoring procedures, the power plant's operating hours will be closely monitored. In the event of a malfunction with any of the monitoring devices, portable instruments will be utilized to ensure that daily monitoring of the deficient parameter(s) is conducted. The recorded data will be documented on paper to maintain accuracy. Biotrend, in accordance with the provider recommendations, has established a set of emergency procedures.

